Technical Appendix

A Construct Ledger for Organizational Science

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Repository and live artifact: https://ccfarre.github.io/construct-ledger/

This appendix records what was run, in what order, with what settings, and what came out. Every number in the proposal appears here with the procedure that produced it. The tools are in `tools/` in the repository and each one takes `--selftest`, which runs without network access and without spending anything.

A. Corpus construction

Source. OpenAlex for article metadata and inverted abstracts, Crossref for journal identity. Journals are resolved to ISSNs through Crossref before any records are pulled, so membership is decided by identifier rather than by title string.

Journal	Publisher	ISSN (print, electronic)
Academy of Management Journal	Academy of Management	0001-4273, 1948-0989
Academy of Management Review	Academy of Management	0363-7425, 1930-3807
Administrative Science Quarterly	SAGE	0001-8392, 1930-3815
Organization Science	INFORMS	1047-7039, 1526-5455
Strategic Management Journal	Wiley	0143-2095, 1097-0266

Window. 1990 to 2025 inclusive. Abstracts are reconstructed from OpenAlex's inverted index.

Inclusion rules and attrition. Every excluded record carries a machine-readable reason, written to `corpus_mgmt.attrition.csv` as the pull runs.

Reason for exclusion	Records
No abstract deposited	1,983
Abstract shorter than 40 words	1,057
No DOI, so journal membership cannot be verified	192
Editorial, erratum or other paratext by OpenAlex type	8
Abstract longer than 600 words	6
Retained	**9,111**

The 40-word floor removes one-line publisher stubs; the 600-word ceiling removes records where a full-text first page has been deposited in the abstract field. Both thresholds are arguments to the tool and are recorded in the run manifest.

Coverage caveat. Deposit practice varies by publisher and era. Effective coverage against publisher-deposited article counts is close to complete for SMJ, Organization Science and AMJ, and roughly 37 percent for ASQ, whose older volumes have no deposited abstracts. The ledger therefore describes the record as deposited, not the journals' full output, and ASQ is under-represented in consequence.

B. Cleaning

Abstracts carry boilerplate that is not part of the abstract. Ten trailing rules remove copyright lines, publisher notices, PsycINFO trailers, open-access licence text and platform use notices; three leading rules remove "Abstract" labels and summary headings.

Two guards exist because the first version of the cleaner damaged text. Rules match from their marker to the end of the string rather than to the next full stop, because full stops inside licence URLs truncated an Organization Science abstract in the middle of a word. A rule is then refused unless it starts at a word boundary, begins within 300 characters of the end, and removes less than 25 percent of the abstract.

2,812 of 9,111 records were changed. The most frequent rule combinations were a leading "Abstract" label with a copyright notice (867), a leading label with summary headings (598), and a copyright notice alone (481). Nine records were flagged for a human glance rather than changed, because they contain an email address.

Three records were marked not-abstracts and excluded, leaving **9,108 usable**: two ASQ records from 1993 that are JSTOR page headers carrying title, author and citation metadata, and one SMJ record that fell under 40 words once its boilerplate was removed. Every change is logged per document in `clean_report.csv` with the rules applied and the word count before and after.

C. Extraction

Model and settings. `gemini-3.6-flash`, temperature 0, `responseMimeType: application/json`, 4,096 maximum output tokens. The API key travels in the `x-goog-api-key` header; a selftest asserts it never appears in a URL.

System prompt. The model is told it is building an auditable ledger, that it extracts only what the text supports, that it never invents a construct and never imports one from background knowledge the abstract does not theorise. A construct is defined as an abstract theoretical concept the paper reasons with, and explicitly not a method, dataset, firm, industry, country, statistical technique or generic research noun.

User prompt, version a. Asks for a normalised name of at most six words, the definition as this paper uses it in one sentence, the structural role it plays in this paper's argument, and a provenance span of at least three words copied verbatim from the abstract. Journal, year and title are supplied as context; the abstract is delimited by triple quotes.

Schema, version min. Each construct carries `name`, `definition`, `structural_role` from the closed set {antecedent, outcome, mediator, moderator, focal, context}, and `provenance_span`. Relationships between constructs are requested in the same object and retained in the ledger but not used in the analyses reported here.

Validation, applied to every returned object. A construct is dropped if it has no name, if the name is a generic research noun from a fixed list of eighteen, if the name exceeds six words, if the provenance span is shorter than three words, or if the span does not occur verbatim in the abstract after whitespace normalisation. An unrecognised structural role is blanked rather than dropped.

Repair loop. When validation reports problems, the specific problems are fed back with the previous answer and the abstract, and the model is asked to fix those problems, to drop any construct it cannot ground rather than invent a span, and to keep every construct that was already fine. Up to two repairs are allowed. Feeding the errors back rather than resampling is what makes the loop converge: on a paired 195-document comparison it raised the grounded yield from 11 grounding failures and 5 document failures to zero of each, while changing constructs per document only from 4.49 to 4.45.

Attempts across the final ledger of 9,086 documents.

Attempts	Documents	Share
1	8,389	92.3%
2	644	7.1%
3	53	0.6%

Residual problems. Five documents reached the attempt limit with a problem the loop could not resolve. In each case the offending construct was dropped and the document's remaining constructs were kept. Three were names longer than six words; two were grounding problems, one span not found verbatim and one span shorter than three words. Those two are the only constructs in the entire run that the model proposed and could not ground after three attempts, out of 40,981 proposed, and neither entered the ledger.

Refusals. 22 documents returned no construct at all. Every one is a record whose deposited text is not an abstract: eight AMR "Publications Received" book lists, an editor's comment, an obituary, an SMJ erratum notice, an editorial-team announcement, a presidential address, two AMR dialogue pieces, and nine cases where the publisher deposited an acknowledgements footnote, a JSTOR rights notice, an epigraph or an author list instead of the abstract. The accounting closes exactly: 9,086 extracted plus 22 refused equals 9,108 usable. All 22 are listed with their reason in `failures.jsonl`.

Yield. 40,979 construct mentions, a mean of 4.51 per document, median 4, fifth percentile 2, ninety-fifth percentile 7, maximum 13. No document in the ledger has an empty construct list and no `doc_id` appears twice.

Structural roles. antecedent 12,152; outcome 11,515; focal 6,018; moderator 4,682; mediator 4,648; context 1,449; absent or unrecognised 515 (1.3%).

D. Grounding verification

The verification is independent of the extractor. It reads the ledger and the cleaned corpus, normalises whitespace and case, and asks whether each construct's span occurs as a substring of its own abstract.

Result: 40,979 of 40,979 spans found verbatim, grounding rate 1.0000. No span is shorter than three words. The check is also recomputed inside `build-assets-mgmt` when the site data is built, so the figure shipped with the artifact is produced by a second implementation rather than copied from the first.

This is the measurement that makes the rest of the ledger usable. It is a string match, it costs nothing, it needs no API key, and it will still run when the model that produced the ledger has been retired.

E. Canonicalisation

Embedding. `gemini-embedding-001`, task type `SEMANTIC_SIMILARITY`, 3,072 dimensions, over the distinct `name: definition` strings. The management corpus has 40,945 such strings from 40,979 mentions. The store is compacted to float32 at 768 leading dimensions for all downstream work, which the model's Matryoshka property supports.

Calibration. Positives are pairs of distinct wordings that the extractor independently gave the same normalised name; negatives are pairs drawn from different names. 3,000 pairs of each, seed 17. The reported threshold is the lowest value at which false merges among negatives fall below one percent.

Threshold	Management: same-construct retained	Management: false merges	IS: same-construct retained	IS: false merges

0.78	100.0%	21.27%	99.9%	18.73%
0.80	99.8%	6.70%	99.8%	6.53%
0.82	98.9%	1.80%	99.0%	2.33%
0.84	**95.5%**	**0.37%**	**97.4%**	**0.57%**
0.86	87.7%	0.17%	92.8%	0.10%
0.88	72.9%	0.07%	83.3%	0.03%
0.90	52.9%	0.00%	67.9%	0.00%

The one-percent point is 0.83 for management and 0.84 for information systems. Similarity distributions are close across the two fields: random-pair medians of 0.756 and 0.751, same-name medians of 0.903 and 0.920. The ledger uses **0.84 for both**, the stricter of the two, so that the fields are compared at one value rather than at two tuned separately.

Merge rule. Two conditions must both hold. Cosine similarity between the two groups' mean embeddings must reach the threshold, and one name's token set must be a superset of the other's after stopword removal and plural stemming, with a single-token name never absorbed into a longer one. The second condition is a restriction, not a licence: removing it merges names that share only a head noun, which is why it exists. Candidate pairs are restricted by blocking on discriminating tokens, defined as tokens appearing in at most two percent of groups, which is what makes a recall-friendly threshold computationally safe.

Result. 40,979 mentions, 31,786 groups after exact-name merging, 23,522 canonical constructs after semantic merging. 18,896 appear in a single paper; 1,083 appear in five or more. 469 head-noun families contain at least one construct used in five or more papers.

The same pipeline over 8,424 information systems abstracts from seven IS journals yields 29,387 mentions and 16,426 constructs, which reproduces the published figures for that corpus exactly.

F. Threshold sensitivity

Threshold	Constructs	Single-mention share
0.83	23,333	79.3%
0.84	23,522	79.9%
0.88	25,370	84.2%

Moving the threshold from 0.83 to 0.88 changes the construct count by 8.7 percent. Of the 2,118 groups that lead at 0.88 and are absorbed at 0.83, **1,968 (92.9 percent) have a single mention**. Constructs with ten or more mentions move from 401 to 436 across the same range. The implication is specific rather than general: statistics on the single-use tail are threshold-dependent and should be reported with the sensitivity attached, while statistics on the recurring core are stable.

G. Known failure modes

The qualification rule over-merges. The rule was designed for a field where the modifier individuates the construct, so that "user satisfaction with MDS" is a usage of "user satisfaction". In management the shorter name is often the general construct and the longer one a theorised specialisation, and the rule absorbs the distinction. Observed at 0.84: *transient competitive advantage* into *competitive advantage*, *sustainable competitive advantage* into *competitive advantage*, *firm specific human capital* into *human capital*, *corporate social performance* into *corporate performance*, *strategic decision making* into *decision making*. The same shape occurs in information systems, where *IT innovation* and *IS innovation* are absorbed into *innovation*, so

this is a property of the rule and not of either field. It biases the management count downward, which is conservative with respect to any claim that management carries more constructs.

A second kind appears in the same band: names that denote a process, rate or property of a construct rather than the construct itself, such as *erosion of competitive advantage*, *human capital losses* and *team performance variability*. These are absorbed for the same reason. Distinguishing them from genuine one-off phrasings cannot be done from the names alone, which is why every merge is published rather than summarised.

The instrument shapes the count. The IS ledger was extracted with the fuller schema, which also requests compositional origin and artifact proximity, and yields 3.49 mentions per abstract against 4.51 for management. Until one prompt is run over both corpora, document-level comparisons are confounded. Mention-normalised comparisons are unaffected: constructs per mention are 0.559 for IS and 0.574 for management, and single-use shares are 76.3 and 79.9 percent.

Abstracts are not papers. A construct introduced in the theory section and never named in the abstract is invisible. The ledger measures the constructs a paper advertises, which is a defensible object of study but not the same as every construct a paper uses.

Model dependence. Prompt version, schema version, model name and repair budget are recorded on every ledger row as a single version key, in this case `gemini-3.6-flash/a/min/r2`. The canonicaliser refuses to merge across different version keys.

H. Cost and compute

Costs are metered from the token counts the API returns, summed over the final ledger, at $0.30 per million input tokens and $4.00 per million output tokens.

Step	Cost	Wall clock	Concurrency
Corpus assembly	free	41 min	serial
Cleaning	free	under 1 min	serial
Extraction, 9,086 abstracts	**$12.49**	206 min	8 workers
Embedding, 40,945 strings	about $0.20 (estimated)	15 min	4 workers
Compaction, calibration, merging	free	4 min	serial
Grounding verification	free	seconds	serial
Site asset build	free	2 min	serial
Total	**about $12.69**	**about 4.5 hours**	

Measured token use is 5,179,226 input and 2,734,620 output, a mean of 570 in and 301 out per document. The embedding figure is estimated from character counts because that endpoint does not return token counts.

Every cost-bearing tool prints the estimate and refuses to send anything without `--go`, meters spend as it runs, stops at a ceiling computed from observed rather than assumed cost per document, and never re-pays for work already in the output file. The ceiling guard runs one document before opening the worker pool, so the exposure before any cost is observed is one document rather than the full pool.

I. Shipped data

```
data/mgmt/meta.json          corpus counts, calibration, grounding rate
data/mgmt/constructs.json    one record per construct: name, key, mentions, papers,
                             first and last year, roles, surface-form count, head-noun family
data/mgmt/figures.json       precomputed series behind the figures
```

```
data/mgmt/graph.json         co-occurrence ties, construct neighbourhoods, head-noun families
data/mgmt/detail/NNNN.json   per construct: up to five materially distinct definitions with
                             year, journal and DOI; up to six provenance records; the names
                             merged into it; its journals, years and role counts
```

Two constructs are tied when a paper asserts both. A raw count ties everything to the most common constructs, so a tie is kept only when the pair co-occurs at least twice and at least fifty times more often than their separate frequencies predict. That leaves 899 ties over 951 constructs and 107 neighbourhoods of three or more, which read as literatures: transaction-cost economics and alliances, micro organizational behaviour, teams, upper echelons and governance, diversification economics, the resource-based view, contingency theory. Head-noun families are indexed over all 23,522 constructs, 4,095 families in total.

Placing a construct the ledger has never seen needs more than its head noun, because a construct nobody has used has no co-occurrence ties by definition. The interface therefore applies the ledger's own rule to the text in front of it: two constructs are tied when they share a sentence in the submitted manuscript, exactly as two constructs are tied when they share an abstract in the corpus. A new construct then enters the corpus structure through the constructs it keeps company with in its author's own prose. On a 15,000-word manuscript tested during development, seven of the eight candidate constructs the ledger had not seen were tied in this way, and the eighth was reported as isolated rather than silently dropped.

Definitions are kept only when they differ from those already retained, measured as a Jaccard overlap below 0.55 on content tokens, and are sampled across the construct's full period rather than taken from its earliest mentions. Provenance records are sampled the same way. Each carries the span, its character offsets, 120 characters of context either side, and the DOI.

J. Reproduction

```
tools/iscorpus-mgmt-v2 resolve --journals journals.json
tools/iscorpus-mgmt-v2 pull    --out corpus_mgmt.jsonl --from 1990 --to 2025 --go
tools/isclean-mgmt             --corpus corpus_mgmt.jsonl --out corpus_mgmt_clean.jsonl \
                               --report clean_report.csv
tools/iscx-mgmt-v4             --corpus corpus_mgmt_clean.jsonl --out ledger_mgmt_v2.jsonl \
                               --failures failures.jsonl --workers 8 --max-cost 13 --go
tools/iscledger-mgmt           --ledger ledger_mgmt_v2.jsonl --db mgmt_ledger.sqlite
tools/iscembed-mgmt            --ledger mgmt_ledger.sqlite --out emb_mgmt.sqlite --workers 4 --go
tools/iscgeo compact           --emb emb_mgmt.sqlite --out emb_mgmt_fast.sqlite --dims 768
tools/iscgeo calibrate         --ledger mgmt_ledger.sqlite --corpus corpus_mgmt_clean.jsonl \
                               --emb emb_mgmt_fast.sqlite --dims 768
tools/iscgeo canon             --ledger mgmt_ledger.sqlite --corpus corpus_mgmt_clean.jsonl \
                               --emb emb_mgmt_fast.sqlite --dims 768 --threshold 0.84 \
                               --out canon_mgmt_t84.jsonl
tools/build-assets-mgmt        --canon canon_mgmt_t84.jsonl --ledger ledger_mgmt_v2.jsonl \
```

```
--corpus corpus_mgmt_clean.jsonl --out data/mgmt \
--label "Management and organization theory" --threshold 0.84
```

An OpenAlex API key is required from 13 February 2026 and a Gemini key for the two paid steps. Both are read from the environment and neither is written to any file. Python standard library only, except `numpy` for the merge step. Everything after extraction and embedding is deterministic and free, so a reader who takes the shipped ledger can reproduce every canonicalisation and every figure at no cost.

K. Ethics and redistribution

Abstracts are the publishers' and are not redistributed. The repository ships each provenance span with 120 characters of context and the DOI, which is what a reader needs to verify a given extraction and nothing more. Construct names, definitions and structural roles are model output over published abstracts. No human subjects are involved and no personal data is processed. The repository is MIT for code; the derived ledger is released for research use.

The one judgment worth stating plainly: a ledger of this kind can be read as a verdict on individual papers, and it is not one. A construct appearing once in the corpus is not thereby a bad construct, and a construct appearing in two hundred papers is not thereby a good one. The ledger measures usage, and usage is evidence about a field rather than about any paper in it.